

CPC Zurich 2026 Practical Tutorial

Reinforcement Learning using hBayesDM

Installation Guide

Updated by Dongwon Lee (dongwon99@snu.ac.kr),

Authors / Tutors: Woo-Young Ahn (wahn55@snu.ac.kr), Dongwon Lee (dongwon99@snu.ac.kr), and Jinwoo Jeong (jwjeong96@gmail.com),
Computational Psychiatry Course 2026, Zurich, Switzerland.

This tutorial will make use of the R version of the hBayesDM package. There are two parts in this installation guide, one dedicated to the installation on **Windows** (part A) and the other to the installation on **macOS** (part B).

NOTE: It may take some time to install the files (about 10 mins), so we recommend you install in advance

PART A) How to install R version hBayesDM on Windows:

1. Install R (version 4.4.0 or later)

Download and install the latest version of R compatible with your operating system:
<https://cran.r-project.org/>

2. Install RStudio Desktop

Download and install the latest version of RStudio Desktop:
<https://posit.co/download/rstudio-desktop/>

3. Install Rtools

Download and install the version of Rtools compatible with your version of R:

<https://cran.r-project.org/bin/windows/Rtools/>

Rtools provides the C++ compiler and build utilities required to compile Stan models on Windows.

4. Install CmdStanR and CmdStan

Install the latest version of CmdStanR package by typing the following command in R studio Console:

```
install.packages(  
  "cmdstanr",  
  repos = c(  
    "https://stan-dev.r-universe.dev",  
    getOption("repos")  
  )  
)
```

Verify that the C++ toolchain is configured correctly:

```
cmdstanr::check_cmdstan_toolchain()
```

Then, install CmdStan:

```
cmdstanr::install_cmdstan()
```

5. Install hBayesDM

hBayesDM can be installed either from CRAN or from the official GitHub repository.

Option 1. Install hBayesDM from CRAN:

```
install.packages("hBayesDM")
```

Option 2. Install hBayesDM from the official GitHub repository:

Install remotes if it is not already installed

```
install.packages("remotes")
```

Check the installed remotes version

```
packageVersion("remotes")
```

Install hBayesDM from the official GitHub repository

```
remotes::install_github(  
  "CCS-Lab/hBayesDM",  
  ref = "v2.0.1",  
  subdir = "R"  
)
```

6. Verify that the following test code runs without any error:

```
library(hBayesDM)  
  
# Test if the package works fine using example data  
output1 <- gng_m1(  
  data="example",  
  niter=2000,  
  nwarmup=1000,  
  nchain=2,  
  ncore=2)  
  
# plot the output
```

```
plot(output1)
```

For more information, please check this tutorial:

https://ccs-lab.github.io/hBayesDM/articles/getting_started.html

Note: The first run of a model may take additional time because the Stan model needs to be compiled. Subsequent runs can reuse the compiled model.

PART B) How to install R version hBayesDM on macOS:

1. Install R (version 4.4.0 or later)

Download and install the latest version of R for macOS:

<https://cran.r-project.org/bin/macosx/>

2. Install RStudio Desktop

Download and install the latest version of RStudio Desktop:

<https://posit.co/download/rstudio-desktop/>

3. Install Xcode Command Line Tools

CmdStan requires a C++ compiler and build tools to compile Stan models on macOS.

Open Terminal and run:

```
xcode-select --install
```

Follow the on-screen instructions to complete the installation.

4. Install CmdStanR and CmdStan

Install CmdStanR by typing the following command in the RStudio Console:

```
install.packages(  
  "cmdstanr",  
  repos = c(  
    "https://stan-dev.r-universe.dev",  
    getOption("repos")  
  )  
)
```

Verify that the required C++ toolchain is configured correctly:

```
cmdstanr::check_cmdstan_toolchain()
```

Install CmdStan:

```
cmdstanr::install_cmdstan()
```

5. Install hBayesDM

hBayesDM can be installed either from CRAN or from the official GitHub repository.

Option 1. Install hBayesDM from CRAN:

```
install.packages("hBayesDM")
```

Option 2. Install hBayesDM from the official GitHub repository:

Install remotes if it is not already installed

```
install.packages("remotes")
```

Check the installed remotes version

```
packageVersion("remotes")
```

Install hBayesDM from the official GitHub repository

```
remotes::install_github(  
  "CCS-Lab/hBayesDM",  
  ref = "v2.0.1",  
  subdir = "R"  
)
```

6. Verify that the following test code runs without any error:

```
library(hBayesDM)  
  
# Test if the package works fine using example data  
output1 <- gng_m1(  
  data="example",  
  niter=2000,  
  nwarmup=1000,  
  nchain=2,
```

```
ncore=2)  
  
# plot the output  
plot(output1)
```

For more information, please check this tutorial:

https://ccs-lab.github.io/hBayesDM/articles/getting_started.html

Note: The first run of a model may take additional time because the Stan model needs to be compiled. Subsequent runs can reuse the compiled model.